



The 1983 Hartle–Hawking Paper: Mathematical Framework

The 1983 paper "Wave function of the Universe" (*Phys. Rev. D* 28, 2960) defines a wave function Ψ for the universe via a Euclidean gravitational path integral over compact four-geometries with a single boundary, subject to the no-boundary condition.^[1]

Gravitational Action

The starting point is the Einstein–Hilbert action with cosmological constant Λ and matter fields ϕ , with the Gibbons–Hawking–York boundary term included :

$$S = \int_{\mathcal{M}} d^4x \sqrt{-g} \left(\frac{R}{2} - \Lambda + \mathcal{L}_m \right) + \int_{\partial\mathcal{M}} d^3y \sqrt{h} K$$

where R is the four-dimensional Ricci scalar, h_{ij} is the induced three-metric on the boundary $\partial\mathcal{M}$, and K is the trace of the extrinsic curvature. The GHY surface term eliminates second derivatives from the variational problem, enabling Dirichlet boundary conditions (fixed boundary metric).

ADM Decomposition and the Wheeler–DeWitt Equation

The four-metric is decomposed via the ADM formalism:

$$ds^2 = -(N^2 - N_i N^i) dt^2 + 2N_i dx^i dt + h_{ij} dx^i dx^j$$

where N is the lapse, N_i the shift vector, and h_{ij} the spatial three-metric. The lapse and shift are Lagrange multipliers enforcing the Hamiltonian and momentum constraints. Canonical quantisation replaces the momentum conjugate to h_{ij} with a functional derivative, yielding the Wheeler–DeWitt equation:^[2]

$$\hat{H} \Psi[h_{ij}, \phi] = 0$$

This equation has no explicit time variable — the wave function depends only on the three-geometry and matter configuration on a spatial slice, not on any external time parameter. The no-boundary proposal selects a particular solution to this otherwise underdetermined equation.^[2]

The No-Boundary Wave Function via Euclidean Path Integral

The central definition in the paper is the wave function computed by a Euclidean (Wick-rotated) Feynman path integral :

$$\Psi[h_{ij}, \phi_0] = \int \mathcal{D}g_{\mu\nu} \mathcal{D}\phi e^{-I_E[g_{\mu\nu}, \phi]/\hbar}$$

where I_E is the Euclidean action (obtained from S by the rotation $t \rightarrow -i\tau$), and the integral is taken over all compact, positive-definite four-geometries $g_{\mu\nu}$ and regular matter configurations ϕ that:

- are compact (no boundary to the past),
- induce the three-metric h_{ij} and field value ϕ_0 on the single boundary $\partial\mathcal{M}$.^[3]

The Euclidean action for pure gravity with cosmological constant is:

$$I_E = - \int_{\mathcal{M}} d^4x \sqrt{g_E} \left(\frac{R_E}{2} - \Lambda \right) - \int_{\partial\mathcal{M}} d^3y \sqrt{h} K$$

with the sign flipped relative to the Lorentzian action, so that the weighting $e^{-I_E/\hbar}$ suppresses geometries with large Euclidean action .

Minisuperspace Reduction

The paper restricts to the closed FLRW minisuperspace with metric :

$$ds^2 = -\tilde{N}^2(t) dt^2 + a^2(t) d\Omega_3^2$$

where $d\Omega_3^2$ is the round metric on S^3 (volume $2\pi^2$) and $a(t)$ is the scale factor. Under this ansatz the action reduces to :

$$S = 2\pi^2 \int_0^1 dt \tilde{N} \left(-3a \frac{\dot{a}^2}{\tilde{N}^2} + 3a - \Lambda a^3 \right)$$

Note the wrong-sign kinetic term $-3a\dot{a}^2/\tilde{N}$, reflecting the conformal-mode instability intrinsic to gravity . A further variable change $q = a^2$ and lapse rescaling renders the action quadratic in q :

$$S = 2\pi^2 \int_0^1 dt_q \left(-\frac{3}{4N} \dot{q}^2 + N(3 - \Lambda q) \right)$$

Path Integral Evaluation

The minisuperspace path integral becomes an ordinary integral over the lapse times a functional integral over q :

$$G[q_1; q_0] = \int_{\mathcal{C}} dN \int_{q(0)=q_0}^{q(1)=q_1} \mathcal{D}q e^{iS(N, q)/\hbar}$$

Since the action is quadratic in q , the functional integral is Gaussian and evaluable exactly. Writing $q(t_q) = \bar{q}(t_q) + Q(t_q)$ where $\bar{q}$ solves the equation of motion, the linear term vanishes and one obtains :

$$G[q_1; q_0] = \sqrt{\frac{3\pi i}{2\hbar}} \int_C \frac{dN}{N^{1/2}} e^{2\pi^2 i S_0/\hbar}$$

with the on-shell action:

$$S_0 = \frac{\Lambda^2}{36} N^3 + \left(3 - \frac{\Lambda}{2}(q_0 + q_1)\right) N - \frac{3(q_1 - q_0)^2}{4N}$$

Saddle Points

The remaining lapse integral is evaluated by saddle-point (steepest-descent) methods. Setting $\partial S_0/\partial N = 0$ yields four saddle points :

$$N_\sigma = c_1 \frac{3}{\Lambda} \left[\left(\frac{\Lambda}{3} q_0 - 1 \right)^{1/2} + c_2 \left(\frac{\Lambda}{3} q_1 - 1 \right)^{1/2} \right], \quad c_1, c_2 \in \{-1, +1\}$$

with on-shell action at the saddles:

$$S_0^{\text{saddle}} = -c_1 \frac{6}{\Lambda} \left[\left(\frac{\Lambda}{3} q_0 - 1 \right)^{3/2} + c_2 \left(\frac{\Lambda}{3} q_1 - 1 \right)^{3/2} \right]$$

For classical boundary conditions ($q_0, q_1 > 3/\Lambda$), these saddles are real and two of them (on the positive real N axis) contribute, yielding a propagator approximated by a sum of two phases corresponding to expanding and bouncing de Sitter geometries .

The No-Boundary Condition Itself

The no-boundary proposal corresponds to taking one boundary to zero size, $q_0 \rightarrow 0$. This is a classically forbidden (sub-barrier) configuration since $q_0 < 3/\Lambda$. The saddle points become complex :

$$N_\sigma = c_1 \frac{3}{\Lambda} \left[i \left(1 - \frac{\Lambda}{3} q_0 \right)^{1/2} + c_2 \left(\frac{\Lambda}{3} q_1 - 1 \right)^{1/2} \right]$$

Only a single saddle contributes (selected by Picard-Lefschetz theory applied to the contour $\mathcal{C} = (0^+, \infty)$), and the wave function takes the form :

$$\Psi[q_1] \approx \exp\left(-\frac{12\pi^2}{\hbar\Lambda} \left(1 - \frac{\Lambda}{3} q_0\right)^{3/2}\right) \exp\left(-i \frac{12\pi^2}{\hbar\Lambda} \left(\frac{\Lambda}{3} q_1 - 1\right)^{3/2}\right)$$

Setting $q_0 = 0$, the real exponential prefactor becomes $\exp(-12\pi^2/\hbar\Lambda)$, a finite amplitude for tunnelling from "nothing" (zero three-geometry) to a finite universe of size q_1 . The associated saddle-point geometry is complex in the interior but smoothly caps off — the Euclidean section

near $q_0 = 0$ is a portion of a round four-sphere S^4 , which has no boundary, replacing the classical big-bang singularity with a regular point analogous to the South Pole of a sphere.^[4]

Physical Content

The real exponential factor $e^{-12\pi^2/\hbar\Lambda}$ controls the relative probability of nucleation: universes with larger Λ are exponentially preferred, which in the inflationary context implies that the no-boundary wave function favours an initial period of inflation. The oscillatory factor encodes a WKB-like classical phase, and the condition that Ψ be well approximated by $e^{iS_{\text{classical}}/\hbar}$ in the large- q_1 regime is the criterion for the emergence of classical spacetime from the quantum state.^{[5] [3]}

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1. https://en.wikipedia.org/wiki/Hartle-Hawking_proposal
2. https://en.wikipedia.org/wiki/Wheeler-DeWitt_equation
3. <https://arxiv.org/html/2303.08802v3>
4. <https://science.howstuffworks.com/dictionary/famous-scientists/physicists/stephen-hawking3.htm>
5. <https://link.aps.org/doi/10.1103/PhysRevLett.100.201301>